

Spring 2026

Math F251X

Calculus 1: Midterm 1

Name: _____

Section: ☐ 9:15am (James Gossell)

☐ 11:45am (Gordon Williams)

☐ async (James Gossell)

Rules:

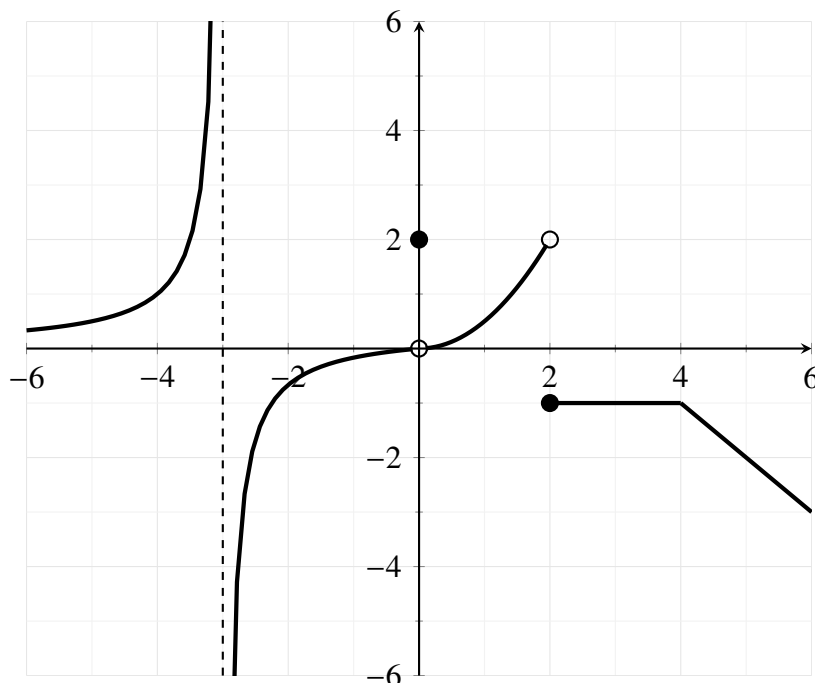
- You will have 90 minutes to take this exam.
- Partial credit will be awarded, but you must show your work.
- You may have a single handwritten $3'' \times 5''$ notecard, both sides.
- Calculators are **not** allowed.
- Place a box around your FINAL ANSWER to each question where appropriate.
- Turn off anything that might go beep during the exam.

Good luck!

Problem	Possible	Score
1	16	
2	8	
3	16	
4	10	
5	10	
6	12	
7	16	
8	12	
Extra Credit	(5)	
Total	100	

1. (16 points)

Use the graph of $f(x)$ to answer the following questions.



- a. Fill in the blanks below. If the value does not exist or is undefined, write DNE.

$$\begin{array}{lll}
 \lim_{x \rightarrow -3^-} f(x) = +\infty & \lim_{x \rightarrow 0} f(x) = 0 & f(0) = 2 \\
 \lim_{x \rightarrow 2^-} f(x) = 2 & \lim_{x \rightarrow 2^+} f(x) = -1 & \lim_{x \rightarrow 2} f(x) = \text{DNE} \\
 f'(3) = 0 & \lim_{x \rightarrow 4^-} f'(x) = 0 & \lim_{x \rightarrow -4^+} f'(x) = -1
 \end{array}$$

- b. State the x -values for which f is **not continuous**. For each of your answers, classify the discontinuity as **jump**, **removable**, **infinite**, or **other**.

-3 , infinite 0 , removable 2 , jump

- c. State the x -values for which f is **not differentiable** (where $f'(x)$ does not exist).

-3 , 0 , 2 and 4

2. (8 points)

The distance d in centimeters of a ball from a wall at time t is given by $d(t) = 10 + 3 \cos\left(\frac{5\pi}{3}t\right)$, where t is measured in seconds. We are interested in the velocity of the ball after one second.

In the table below, v_{avg} denotes the average velocity of the ball on the time interval between the time 1 second, and the time t seconds. Note that $d(1) = 23/2$.

t sec	$d(t)$ cm	v_{avg} cm/sec
0.00000	13	$-3/2$
0.9	10.	15.
0.99	11.362	13.8029
0.999	11.4864	13.624
0.9999	11.4986	13.6056
0.99999	11.4999	13.6037
1.00001	11.5001	13.6033
1.0001	11.5014	13.6014
1.001	11.5136	13.5829
1.01	11.6339	13.3917
1.1	12.5981	10.9808
2	17/2	-3

$$d(0) = 10 + 3 \cos(0) = 13 \text{ sec}$$

$$v_{\text{avg}} = \frac{23/2 - 13}{1 - 0} = \frac{-3/2}{1} = -3/2 \text{ cm/sec}$$

- a. Fill in the missing entries in the table above (for $t = 0.00000$ seconds).

- b. Using the data provided in the table above, estimate a value for the instantaneous velocity of the ball when $t = 1$ second.

$$v_{\text{inst}} \approx 13.6035 \text{ cm/sec}$$

- c. Is the ball moving towards the wall, or away from the wall when $t = 1$ second?

away, since $v_{\text{inst}} > 0$.

3. (16 points)

Evaluate the following limits. Show your work. Use limit notation where necessary; you will be graded both on your computation and on your correct use of notation. If the limit does not exist, write DNE and a few words about why it does not exist. If the limit is unbounded, write ∞ or $-\infty$ as appropriate, and again, justify your answer with a few words.

$$\begin{aligned} \text{a. } \lim_{x \rightarrow -3} \frac{\sqrt{x+4} - 1}{x+3} &= \frac{\sqrt{-3+4} - 1}{-3+3} = \frac{0}{0} \\ &= \lim_{x \rightarrow -3} \frac{\sqrt{x+4} - 1}{x+3} \cdot \frac{\sqrt{x+4} + 1}{\sqrt{x+4} + 1} = \lim_{x \rightarrow -3} \frac{(x+4) - 1}{(x+3)(\sqrt{x+4} + 1)} \\ &= \lim_{x \rightarrow -3} \frac{x+3}{(x+3)(\sqrt{x+4} + 1)} = \lim_{x \rightarrow -3} \frac{1}{\sqrt{x+4} + 1} = \frac{1}{2}. \end{aligned}$$

$$\begin{aligned} \text{b. } \lim_{t \rightarrow 2} \frac{\frac{1}{2} - \frac{1}{t}}{t-2} &= \frac{0}{0} \\ &= \lim_{t \rightarrow 2} \frac{1}{t-2} \cdot \left(\frac{t}{2t} - \frac{2}{2t} \right) = \lim_{t \rightarrow 2} \frac{t-2}{2t(t-2)} = \lim_{t \rightarrow 2} \frac{1}{2t} = \frac{1}{4} \end{aligned}$$

$$\text{c. } \lim_{\theta \rightarrow 0} \frac{2\theta}{\cos(\theta)} = \frac{2 \cdot 0}{1} = 0$$

$$\begin{aligned} \text{d. } \lim_{x \rightarrow 1^+} \frac{2xe^{-x}}{x^2(1-x)} & \quad \text{Numerator goes to } 2e^{-1}, \text{ denominator is negative,} \\ & \quad \text{and goes to zero. Hence,} \\ \lim_{x \rightarrow 1^+} \frac{2xe^{-x}}{x^2(1-x)} &= -\infty. \end{aligned}$$

4. (10 points)

Consider the function

$$f(x) = x^2 + 3x.$$

Find $f'(-2)$ using the **limit definition of the derivative** and show your work using all appropriate notation. **No credit will be awarded for using other methods.** Begin by writing down the limit definition of the derivative. You must write limits where necessary to receive full credit.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} [(x+h)^2 + 3(x+h) - [x^2 + 3x]] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} [\cancel{x^2} + 2\cancel{x}h + h^2 + \cancel{3x} + 3h - \cancel{x^2} - \cancel{3x}] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} [2xh + h^2 + 3h] = \lim_{h \rightarrow 0} (2x + h + 3) \\ &= 2x + 3. \end{aligned}$$

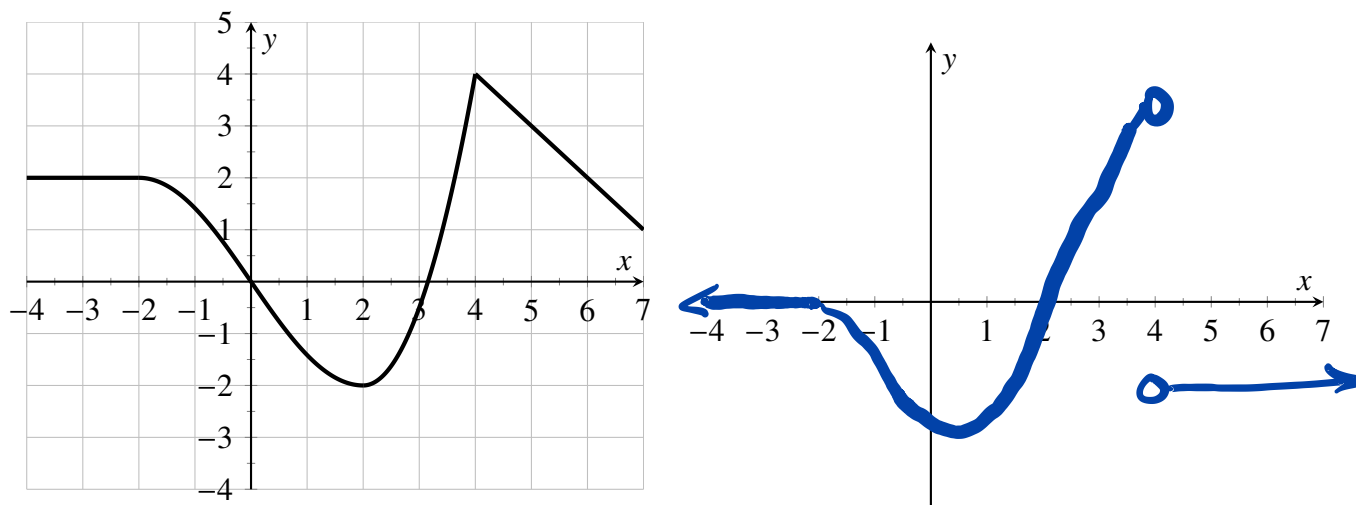
$$\text{So } f'(-2) = -4 + 3 = -1.$$

OR

$$\begin{aligned} f'(-2) &= \lim_{h \rightarrow 0} \frac{f(-2+h) - f(-2)}{h} = \lim_{h \rightarrow 0} \frac{(-2+h)^2 + 3(-2+h) - ((-2)^2 + 3(-2))}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{4} - 4h + h^2 - \cancel{6} + 3h - \cancel{4} + \cancel{6}}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 - h}{h} = \lim_{h \rightarrow 0} (h - 1) = -1. \end{aligned}$$

5. (10 points)

The graph of $f(x)$ is shown below. On the **other set of axes**, sketch the graph of $f'(x)$. If there are any asymptotes, draw them with dashed lines. Use open circles to show points where the derivative is not defined, if any. (You are not given values on the y-axis; I am interested in the correct shape/holes/asymptotes of the derivative, not the specific values.)



6. (12 points)

Find the derivatives of the following functions. Use appropriate derivative notation (like " $f'(x) =$ ").

a. $f(x) = x^4 + 4x + \frac{1}{x^4} + \cos\left(\frac{\pi}{4}\right)$

$$f'(x) = 4x^3 + 4 - 4x^{-5}$$

b. $g(x) = \frac{x^2 \sin(x)}{2x-1}$

$$g'(x) = \frac{p'q - pq'}{q^2} = \frac{(2x \sin(x) + x^2 \cos(x)) \cdot (2x-1) - 2 \cdot x^2 \sin(x)}{(2x-1)^2}$$

c. $h(x) = \sqrt{x}(x^2 - 7x + 12) = x^{5/2} - 7x^{3/2} + 12x^{1/2}$

$$h'(x) = \frac{5}{2}x^{3/2} - \frac{21}{2}x^{1/2} + 6x^{-1/2}$$

7. (16 points)

A drone is launched from the ground. Its **height** (in feet) at time t seconds is given by the function

$$h(t) = \frac{32t}{t+1}.$$

a. What is the **initial velocity** of the drone?

$$h'(t) = \frac{32(t+1) - 32t}{(t+1)^2} = v(t)$$

$$h'(0) = \text{initial velocity} = \frac{32-0}{1^2} = 32 \text{ ft/sec.} = v(0)$$

b. After 3 seconds, what is the **velocity** of the drone?

$$h'(3) = \frac{32(4) - 32 \cdot 3}{4^2} = \frac{32}{16} = 2 \text{ ft/sec} = v(?)$$

c. Someone has determined that the acceleration of the drone at time t seconds is given by

$$a(t) = -\frac{64}{(t+1)^3}.$$

Describe in a sentence how they obtained the acceleration from $h(t)$.

they took the derivative with respect to t twice.

d. Is the drone speeding up or slowing down when $t = 3$ seconds? Explain your reasoning.

$$a(3) = \frac{-64}{(3+1)^3} < 0, \text{ while } v(3) = 2 > 0, \text{ so}$$

slowing down.

8. (12 points)

A machine extrudes a metal onto a conveyor belt. Let $M(l)$ denote the amount of mass measured in kilograms extruded after the machine has travelled l meters along the belt.

- a. What does $\frac{M(b) - M(a)}{b - a}$ for $0 < a < b$ measure, and what are the units?

It measures the average mass per meter deposited between position a and b on the conveyor. The units are $\frac{\text{kg}}{\text{m}}$.

- b. What does $M'(l)$ measure, and what are the units?

This is the instantaneous rate of change of mass with respect to the distance traveled. It is also measured in kg/m .

- c. Suppose that $M(4) = 152$, and $M'(4) = 13$. Estimate $M(5)$. Be sure to include appropriate units in your answer.

$$M(5) \approx M(4) + M'(4) \cdot 1\text{m} = 152\text{ kg} + 13\text{ kg/m} \cdot \text{m} = 165\text{ kg}.$$

- d. Can $M'(l)$ be negative? Explain your reasoning.

No. That would mean the extruder was taking mass off the conveyor belt as it traveled.

9. (Extra Credit: 5 points)

Suppose that

$$g(x) = \begin{cases} 4x - x^2 & x \leq 3 \\ 9 - 2x & x > 3 \end{cases}.$$

Is the $g(x)$ differentiable at $x = 3$? (In other words, does $g'(3)$ exist?) If so, what is $g'(3)$?

Either way, justify your conclusions.

$$g(3) = 12 - 9 = 3, \quad \lim_{x \rightarrow 3^+} g(x) = 9 - 6 = 3.$$

$$\lim_{x \rightarrow 3^-} g(x) = 12 - 9 = 3$$

So continuous at $x = 3$.

$$\lim_{x \rightarrow 3^-} g'(x) = \lim_{x \rightarrow 3^-} (4 - 2x) = 4 - 6 = -2$$

$$\lim_{x \rightarrow 3^+} g'(x) = \lim_{x \rightarrow 3^+} (-2) = -2.$$

So yes, $g'(3) = -2$.